CMSC 104 Section 2
Spring 2026
Classwork 8

Object-oriented coding

In-class date:

Officially, May 4

Due Date:

Monday, May 11 before midnight

Assignment:

Create a simple bank account management application which allows a bank manager to create new accounts and to perform certain typical actions. Specifically, your program must allow the manager to:

1. Create a new account with an initial balance.
2. View account details, including: account number, account holder’s name, and balance
3. Deposit money into an account
4. Withdraw money from an account, if the account has sufficient funds for the requested withdrawal. If there is not enough money, the withdrawal request should be rejected and an appropriate error message printed.
5. Transfer money between two accounts

Constraints

Your bank customers must be contained and described as a dictionary. The keys of the dictionary are the account numbers; the values are the objects of class BankAccount, which contain account numbers, owner names, and account balances.

You must define a class, BankAccount, which contains methods to implement the functionality described above. When your program begins, create a new, empty dictionary, called something like bank_customers.

When the user selects the option to create a new account, get the owner's name and the initial balance from the user. Generate a new account number (one-up numbers are fine). Call the method that creates a new object. Then add a new entry to the bank_customers dictionary, with the key being the account number of the new account and the value being the object that was created.

Your program must run in a “while” loop, which runs until the user selects the option to “Exit” the program. You will also need to implement supporting functions.

- You should print a menu of options available to the user at the beginning of each loop execution. If the user chooses an invalid option, you must print an error message and display the menu again.

Reminder

Assignments are your own effort. Do not share your code.

Grading Rubric

- Script works as intended - 65 points
- Proper data structures - correct use of a dictionary - 25 points
- Proper definition of the class to implement the bank functionality - 10 points